INITIAL LEARNING [EW6A-EL]

id0517l2119100

Warning

- **When performing the initial learning, apply the parking brake securely and block the front and rear wheels using wheel blocks so that the vehicle does not move. Otherwise, the vehicle may move which could lead to an accident because the gears are forcibly changed with the engine running and the shift lever in the drive position while in the initial learning.**

Introduction

- If the initial learning is not completed, perform the initial learning if any of the following parts is replaced. Otherwise, engine racing may occur or shift shock may increase.
  - — Control valve body
  - — Oil pressure switch
  - — Clutch (low clutch, high clutch, R-3-5 brake, 2-6 brake, and low and reverse brake)
  - — Automatic Transaxle
- There are two procedures for the initial learning; one using the M-MDS and one which does not. Initial learning can be launched easily using the M-MDS.

Using M-MDS

Preparation before servicing

1. Verify initial learning implementation necessity for replacement parts.

Replacement part	Initial learning implementation necessity
Control valve body replacement	Necessary
Oil pressure switch replacement	Necessary
Clutches*1	Necessary
Automatic transaxle replacement	Necessary
ON/OFF solenoid replacement	Unnecessary
Coupler component replacement	Unnecessary

*1 : Low clutch, high clutch, R-3-5 brake, 2-6 brake, and low and reverse brake

2. Apply the parking brake firmly, and set the wheel blocks for both the front and rear wheels.
3. Inspect the ATF level. (See AUTOMATIC TRANSAXLE FLUID (ATF) INSPECTION [EW6A-EL].)
4. Bring the M-MDS inside the vehicle and connect it to the DLC-2.
5. Perform vehicle identification using the M-MDS.
6. Start the engine.
7. After the vehicle is identified, select the following items from initialization of the M-MDS.
   (1) Select “Data logger”.
   (2) Select “Module”.
   (3) Select “TCM”.
8. While checking the PID TFT (ATF temperature), warm-up the engine and allow the ATF temperature to reach **60—110 °C {140—230 °F}**.

Initial learning procedure

1. After the vehicle is identified, select the following items from initialization of the M-MDS.
   (1) Select “Self-test”.
   (2) Select “Module”.
   (3) Select “TCM”.
   (4) Select “Retrieve CMDTCs”
2. Start the engine and wait for **30 s or more**.

Caution

- **When performing the initial learning, initial learning ends for safety purposes when the brake fluid pressure sensor value is the specification or less. When performing the initial learning, be careful of the following.**
  - **— Forcefully depress the brake pedal while initial learning is being implemented. (Brake hydraulic pressure 5 MPa {51 kgf/cm^2, 725 psi} or more)**
  - **— Do not depress the brake pedal after releasing it. Otherwise, the power brake fluid pressure will decrease and weaken the braking force.**

3. Perform the following steps (1) to (5) with the brake pedal depressed.
   (1) Shift the selector lever to the D position.
   (2) When the delete button on the M-MDS DTC screen is pressed and the screen returns to the DTC display screen, shift the selector lever to the M position (manual position) **within 10 s.**

(3) Verify that the engine speed increases from idling.
- If the engine speed does not increase from idling, switch the ignition off and repeat the procedure from Step 2.

Note
- When the engine speed increases from idling, the initial learning is performed.
- The shift position indication and the gear position indication turn off while the initial learning is being implemented.

(4) Verify that the engine speed returns to idle and M1 is displayed after the shift position indication and the gear position indication flashes **2 times**.
- If the shift position indication and the gear position indication do not flash, switch the ignition off and repeat the procedure from Step 2.



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Note
- Because the shift position indication and the gear position indication flash only once, be careful not to overlook it.
- If the initial learning is shut down for any reason while performing the learning, the shift position indication and the gear position indication do not flash.

(5) Shift the selector lever to the P position and turn off the engine.

4. Release the brake pedal.
5. Clear DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
6. Switch the ignition off.
7. Switch the ignition ON (engine off or on).
8. Using the M-MDS, verify that DTC P06B8:00 has not been stored.
   - If DTC P06B8:00 is displayed, switch the ignition off and repeat the procedure from Step 2. (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
   - If any other DTCs are displayed, repair the malfunctioning location according to the applicable DTC troubleshooting. (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].)

Not Using of M-MDS
Preparation before servicing
1. Verify initial learning implementation necessity for replacement parts.

Replacement part	Initial learning implementation necessity
Control valve body replacement	Necessary
Oil pressure switch replacement	Necessary
Clutches*1	Necessary
Automatic transaxle replacement	Necessary
ON/OFF solenoid replacement	Unnecessary
Coupler component replacement	Unnecessary

*1 : Low clutch, high clutch, R-3-5 brake, 2-6 brake, and low and reverse brake

2. Apply the parking brake firmly, and set the wheel blocks for both the front and rear wheels.
3. Inspect the ATF level. (See AUTOMATIC TRANSAXLE FLUID (ATF) INSPECTION [EW6A-EL].)
4. Start the engine.
5. Warm-up the engine and allow the ATF temperature to reach **60—110 °C {140—230 °F}**.

Initial learning procedure
1. Start the engine and wait for **30 s or more**.

Caution
- **When performing the initial learning, initial learning ends for safety purposes when the brake fluid pressure sensor value is the specification or less. When performing the initial learning, be careful of the following.**
  - **— Forcefully depress the brake pedal while initial learning is being implemented. (Brake hydraulic pressure 5 MPa {51 kgf/cm^2, 725 psi} or more)**

— **Do not depress the brake pedal after releasing it. Otherwise, the power brake fluid pressure will decrease and weaken the braking force.**

2. Perform the following steps (1) to (8) with the brake pedal depressed.
   (1) Shift the selector lever to the N position, press the push button start **one time**, and stop the engine.
   (2) Shift the selector lever to the M- position, press the push button start **one time** while holding the lever in the M- position, and wait for **5 s or more** (ignition switch ON (engine off)).
   (3) Shift the selector lever to the N position, press the push button start **one time**, and start the engine.
   (4) Shift the selector lever to the M+ position andwait for **5 s or more** while holding the lever in the M+ position.
   (5) Shift the selector lever to the M- position and wait for **5 s or more** while holding the lever in the M- position.
   (6) Verify that the engine speed increases from idling.
      • If the engine speed does not increase from idling, switch the ignition off and repeat the procedure from Step 1.

   Note
   • When the engine speed increases from idling, the initial learning is performed.
   • The shift position indication and the gear position indication turn off while the initial learning is being implemented.

   (7) Shift the selector lever to the M position (manual position), and verify that M1 is displayed after the shift position indication and the gear position indication flash **2 times**.
      • If the shift position indication and the gear position indication do not flash, switch the ignition off and repeat the procedure from Step 1.



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   Note
   • Because the shift position indication and the gear position indication flash only once, be careful not to overlook it.
   • If the initial learning is shut down for any reason while performing the learning, the shift position indication and the gear position indication do not flash.

   (8) Shift the selector lever to the P position and turn off the engine.
3. Release the brake pedal.
4. Clear DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
5. Switch the ignition off.
6. Switch the ignition ON (engine off or on).
7. Using the M-MDS, verify that DTC P06B8:00 has not been stored. (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
   • If DTC P06B8:00 is displayed, switch the ignition off and repeat the procedure from Step 1.
   • If any other DTCs are displayed, repair the malfunctioning location according to the applicable DTC troubleshooting. (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].)